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***B.Tech. Degree II Semester Regular and I Semester Supplementary
Examination May 2016***

IT/CS//EC/CE/EE/ME/SE 1102 A / 1202 B ENGINEERING PHYSICS
(2015 Scheme)

Time : 3 Hours

Maximum Marks : 60

PART A
(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) For an ordinary source, the coherence time is 10^{-10} sec. Obtain the degree of non-monochromaticity for $\lambda_0 = 5400 \text{Å}$?
- (b) A signal of 1000 mW is injected into a fibre. The out coming signal from the other end is 40 mW. What is the loss in dB?
- (c) Draw the Planes (100) and (010).
- (d) Calculate the distance between two atoms of diamond structure, if the lattice constant of the structure is 5Å .
- (e) Mention any four applications of metallic glasses.
- (f) What are nanocomposites?
- (g) The critical temperature for mercury with isotopic mass 199.5 is 4.185K. What will be its critical temperature when its isotopic mass is increased to 203.4?
- (h) The mean life of a meson is 2×10^{-8} sec. Calculate the mean life of a meson moving with a velocity $0.8c$.
- (i) Calculate the frequency and wavelength of a photon of energy 75 eV.
- (j) Calculate the total absorption coefficient of cinema hall, whose volume is 8000 m^3 and reverberation time required is 1.8 sec.

PART B

(4 × 10 = 40)

- II. (a) A three level laser emits a light of wavelength 5500Å . What will be the ratio of population of upper energy level (E2) to the lower energy level (E1) if the optical pumping mechanism is shut off? (Assume $T = 300 \text{ K}$) (4)
- (b) What is the fundamental principle of a hologram? How is it produced and how is the image constructed from it? (6)

OR

(P.T.O.)

- III. (a) A light ray enters from air to fibre. The refractive index of air is 1. The fibre has refractive index of core is equal to 1.5 and that of cladding is 1.48. Find the critical angle, the fractional refractive index, acceptance angle and numerical aperture? (4)
- (b) Explain in detail, the difference between single mode step index and multimode graded index fibre. (6)
- IV. (a) A monochromatic beam of x-rays of wavelength $1.24 \text{ \AA}$ is reflected by cubic crystal of KCl. Determine the interplanar distance for (100) and (111) planes. (Given density of KCl = 1980 kg/m^3 and molecular weight = 74.5) (4)
- (b) What is the concept of Miller Indices? Derive the formula for the distance between two adjacent planes of a simple cubic lattice. (6)
- OR**
- V. (a) From the following data calculate the wavelength of neutron beam and its speed. Spacing between successive planes is $3.84 \text{ \AA}$, glancing angle 30° and the order of Bragg reflection = 1? (4)
- (b) Explain shape Memory Effect. (6)
- VI. (a) Determine the transition temperature and critical field at 4.2 K for a given specimen of a superconductor if the critical fields are 1.41×10^5 and $4.205 \times 10^5 \text{ amp/m}$ at 1.41 K and 12.9 K respectively. (4)
- (b) Explain the difference between type I and type II superconductors. (6)
- OR**
- VII. (a) Calculate the speed of the electron which has kinetic energy as 1.02 MeV. (Given rest mass energy of the electron = 0.51 MeV)? (4)
- (b) Obtain Einstein's mass-energy relation. (6)
- VIII. (a) An electron is constrained to move in a one dimensional box of length 0.1 nm. Find the first three energy eigen values and the corresponding de-Broglie wavelengths. (3)
- (b) Derive time dependent Schrodinger wave equation. (7)
- OR**
- IX. (a) A hall of floor is $15 \times 30 \text{ m}^2$ along with height of 6 m, in which 500 people occupy upholstered seat and the remainder sit on wooden chairs. Optimum reservation time for orchestral music is 1.36 sec and absorption coefficient per person is 0.44. Calculate the coefficient of absorption to be provided by the walls, floor and ceiling when the hall is fully occupied. (4)
- (b) Describe the construction and working of a piezoelectric oscillator for the production of ultrasonic waves. (6)

